EPSR-test-ex14 Non-cubic (triclinic) simulation box

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the general-cell code of the `epsr` engine [1,2]: a 9-number `.ato` header (nmol and temperature, a b c, and the angles phib thetac phic) instead of the cubic one, so the minimum-image convention, the cell neighbour lists and the density use a non-orthogonal metric. Needed for crystalline or anisotropic boxes [3].

Code exercised

Fortran (original, fortran_src/):

- `ato_read_and_save.f`: `minitc` (general-cell header, cell vectors, `dmax`), `moloutc` (writes the 9-number header)
- `minimagec.f`: `minimagec` (fractional-coordinate minimum image), `minimage`
- `cellneighbourlist.f`: `atomcellno`, `makecellneighbourlist` (cells in the skewed box)

Rust (port, rust/epsr-core/src/):

- `ato.rs`: `State::minitc`, `moloutc`
- `neighbour.rs`: `minimagec`, `State::sync_lattice`, `atomcellno`; `state.rs`: `Lattice`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K, here in a triclinic cell with a = b = c = 33.1213 Å, b at phib = 80° from a in the xy plane, and c tilted by thetac = 10° from z at azimuth phic = 20° (EPSR's cell convention). The cell volume, a·b·c·sin(phib)·cos(thetac)~ 35240 Å³, gives an atomic number density of 0.0908 atoms/Å³ (0.0881 in the cubic box). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5].

Inputs

Copy of EPSR-test with only the first line of the `.ato` replaced by the three-line general-cell header `600 300.0` / `33.1213 33.1213 33.1213` / `80.0 10.0 20.0`; the molecular coordinates are unchanged.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
Two chained passes, the second reading the 9-number header written by `moloutc`. Evidence that the general-cell branch is active: the `.out` of the Fortran reference reports number density 0.90808E-01 (cubic EPSR-test: 0.88070E-01), and the written `.ato` keeps the three-line cell header. All 27 output files are identical after both passes.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.